

Customer Shopping Behaviour Dashboard

– Power BI Project

➤ Executive Summary

This project analyzes customer shopping behavior using Power BI to identify purchasing trends, customer demographics, subscription patterns, payment preferences, and factors influencing sales performance. The dashboard provides interactive visualizations that help businesses understand customer behavior and make data-driven decisions.

The analysis was conducted on a customer shopping dataset containing 3,900 customer records and multiple attributes including age, gender, category preferences, payment methods, shipping types, subscription status, purchase amount, and review ratings.

The dashboard enables stakeholders to monitor key performance indicators (KPIs), identify high-value customer segments, evaluate product category performance, and understand customer purchasing habits.

➤ Business Problem

Retail businesses often struggle to understand:

- Which customer segments contribute the most revenue.
- Which product categories generate the highest sales.
- Whether subscriptions influence customer purchasing behavior.
- Which payment methods customers prefer.
- How demographics affect spending patterns.

Without proper analysis, businesses may miss opportunities for targeted marketing, customer retention, and revenue growth.

➤ Project Objectives

The main objectives of this project were:

1. Analyze customer demographics and purchasing behavior.
2. Identify top-performing product categories.
3. Understand subscription trends among customers.
4. Evaluate spending patterns across age groups.
5. Analyze payment method preferences.

6. Measure the impact of discounts on sales.
7. Provide actionable business insights through an interactive dashboard.

➤ **Dataset Overview**

The dataset contains customer-level transaction and demographic information.

Key Columns –

- Customer ID
- Age
- Gender
- Category
- Purchase Amount (USD)
- Review Rating
- Subscription Status
- Payment Method
- Shipping Type
- Discount Applied
- Frequency of Purchases
- Season
- Location
- Previous Purchases
- Promo Code Used

➤ **Data Preparation**

The following transformations were performed:

Data Cleaning

- Verified data quality.
- Removed inconsistencies and null values where applicable.

Calculated Column Created

Age Group

Customers were segmented into five groups:

- Young (10–25)
- Adult (26–40)
- Middle Age (41–60)
- Senior (60+)

This grouping helped analyze customer behavior across different age segments.

DAX Measures Created

Total Customers -

Count of unique customers.

Average Purchase Amount -

Average spending per customer transaction.

Average Review Rating -

Average customer satisfaction rating.

➤ **Dashboard Overview**

The dashboard consists of KPIs, slicers, and multiple visualizations.

KPI Cards

Total Customers

- 3,900 customers

Average Purchase Amount

- \$59.76

Average Review Rating

- 3.75 / 5

These KPIs provide an overall summary of customer activity and satisfaction.

➤ **Visual Analysis**

Subscription Status Analysis

Insight

- Approximately 73% of customers are non-subscribers.

- Approximately 27% are subscribers.

Business Interpretation

The majority of customers purchase without subscribing. This indicates an opportunity to increase customer retention through subscription programs and loyalty benefits.

➤ **Category Performance Analysis**

Customer Count by Category

Categories analyzed:

- Clothing
- Accessories
- Footwear
- Outerwear

Insight -

Clothing has the highest customer engagement, while Outerwear has the lowest.

Business Interpretation -

Clothing products attract the largest customer base and should remain a strategic focus area.

➤ **Sales by Category**

Insight -

- Clothing generates the highest sales revenue.
- Accessories rank second.
- Outerwear generates the lowest sales.

Business Interpretation -

Inventory and marketing efforts should prioritize Clothing and Accessories categories.

➤ **Age Group Analysis**

Customer Distribution -

The Middle Age (41–60) segment has the highest customer count.

Purchase Amount by Age Group -

The Middle Age group contributes the largest share of revenue.

Business Interpretation -

Middle-aged customers represent the most valuable customer segment and should be targeted through personalized campaigns and premium offerings.

➤ **Gender Analysis**

Insight -

Male customers contribute higher purchase amounts compared to female customers.

Business Interpretation -

Marketing strategies can be tailored to improve engagement among female customers while maintaining loyalty among male customers.

➤ **Payment Method Analysis**

Insight -

Credit Card and PayPal are the most preferred payment methods.

Business Interpretation -

Businesses should ensure smooth payment experiences and promotional offers through these payment channels.

➤ **Seasonal Analysis**

Insight -

Sales are highest during Fall and lowest during Summer.

Business Interpretation -

Seasonal marketing campaigns can be designed to maximize revenue during peak seasons and boost demand during low-performing periods.

➤ **Discount Impact Analysis**

Insight -

Customers who did not receive discounts generated slightly higher total purchase amounts.

Business Interpretation -

Discounts do not necessarily drive higher revenue and should be applied strategically to maximize profitability.

➤ **Business Insights**

1. Clothing is the most popular category in terms of both customer count and revenue generation.
2. Middle-aged customers contribute the highest sales and represent the most valuable customer segment.
3. Most customers are non-subscribers, indicating growth opportunities for membership programs.
4. Credit Card and PayPal are the most preferred payment methods.
5. Customer satisfaction is moderate with an average review rating of 3.75, indicating room for service improvements.
6. Seasonal trends significantly impact purchasing behavior.

➤ **Recommendations**

1. Launch targeted marketing campaigns for the Middle Age customer segment.
2. Promote subscription plans through exclusive discounts and loyalty rewards.
3. Increase inventory investment in Clothing and Accessories categories.
4. Improve customer experience to increase review ratings above 4.0.
5. Create seasonal promotions during Summer to increase sales during low-performing periods.
6. Optimize checkout experiences for Credit Card and PayPal users.

➤ **Conclusion**

The Customer Shopping Behavior Dashboard successfully provides a comprehensive view of customer demographics, purchasing trends, product category performance, and customer preferences.

The analysis reveals that Clothing is the leading category, Middle Age customers generate the highest revenue, and most customers are non-subscribers. These findings can support strategic decisions related to marketing, customer retention, inventory planning, and business growth.

This dashboard demonstrates the practical application of Power BI, DAX, data modeling, and business intelligence techniques to transform raw customer data into actionable business insights.